QIPENG CHEN

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University of Alabama, Tuscaloosa, AL, USA
Ph.D. Student in Educational Research Methods

EDUCATION

University of Alabama — Aug 2024 - Present
Ph.D. Student, Educational Research Methods — Tuscaloosa, AL, USA
Advisor: Prof. Kaiwen Man

Zhejiang Normal University — Sep 2020 - Jun 2023
Master of Education — Jinhua, Zhejiang, China
Advisor: Prof. Peida Zhan

Hunan Normal University — 2014 – 2019
Bachelor of Education — Changsha, Hunan, China

PUBLICATIONS

TOTAL CITATIONS: 75 | H-INDEX: 5

1. **Chen, Q.**, Man, K., Zhang, S., & Harring, J. R. (under review). Multi-source process data boosts Bayesian item compromise detection: Integrating responses, response times, and fixation counts. *Psychometrika*.
2. Zhan, P., **Chen, Q.**, Wang, S., & Zhang, X. (2024). Longitudinal joint modeling for assessing parallel interactive development of latent ability and processing speed using responses and response times. *Behavior Research Methods, 56*, 1656–1677. doi:10.3758/s13428-023-02113-5 *[JCR Q1; 5-year IF: 6.8]*
3. Fu, Y., **Chen, Q.**, & Zhan, P. (2023). Binary modeling of action sequences in problem-solving tasks: One- and two-parameter action sequence model. *Acta Psychologica Sinica, 55*(8), 1383. doi:10.3724/SP.J.1041.2023.01383
4. Zhan, P., **Chen, Q.**, Man, K., & Hao, N. (2026). Implementing cognitive component constraints in the development and validation of short-form Raven's Advanced Progressive Matrices based on item response theory. *Current Psychology*, *45*(4), 418. doi:10.1007/s12144-026-09075-9 *[JCR Q1; *Co-first author]*
5. Chen, H., Lin, Z., **Chen, Q.**, & Zhan, P. (2025). Tracking the development of L2 reading subskills using longitudinal cognitive diagnosis models. *Language Assessment Quarterly, 22*(2), 219–241. doi:10.1080/15434303.2025.2497821 *[JCR Q2]*
6. Zhan, P., Gao, F., & **Chen, Q.** (2025). Key-action coding incorporating misconceptions and its application in diagnostic classification analysis of process data. *Journal of Psychological Science, 48*(2), 481–494. doi:10.16719/j.cnki.1671-6981.20250220
7. Li, Y., **Chen, Q.**, Gao, Y., & Liu, T. (2024). Improvement and application of back random response detection: Based on cumulative sum and change point analysis. *Behavior Re-*

search Methods, 56, 8640–8657. doi:10.3758/s13428-024-02495-0 *[JCR Q1; 5-year IF: 6.8]*

8. Luo, Y., **Chen, Q.**, Chen, J., & Zhan, P. (2024). Development and validation of two shortened anxiety sensitivity index-3 scales based on item response theory. *Humanities and Social Sciences Communications, 11*, 1078. doi:10.1057/s41599-024-03615-z *[JCR Q1; 5-year IF: 3.9]*
9. Fu, Y., Zhan, P., **Chen, Q.**, & Jiao, H. (2024). Joint modeling of action sequences and action time in computer-based interactive tasks. *Behavior Research Methods, 56*, 4293–4310. doi:10.3758/s13428-023-02178-2 *[JCR Q1; 5-year IF: 6.8]*
10. Liu, Y., Zhan, P., Fu, Y., **Chen, Q.**, Man, K., & Luo, Y. (2023). Using a multi-strategy eye-tracking psychometric model to measure intelligence and identify cognitive strategy in Raven's Advanced Progressive Matrices. *Intelligence, 100*, 101782. doi:10.1016/j.intell.2023.101782 *[JCR Q1; 5-year IF: 3.2]*
11. Yu, X., Zhan, P., & **Chen, Q.** (2023). Don't worry about the anchor-item setting in longitudinal learning diagnostic assessments. *Frontiers in Psychology, 14*, 1112463. doi:10.3389/fpsyg.2023.1112463 *[JCR Q1; 5-year IF: 3.7]*

CONFERENCES

1. **Chen, Q.**, Man, K., Zhang, S., & Harring, J. R. (2026, April 8–11). *Multi-source process data boosts item compromise detection*. NCME 2026 Annual Meeting, Los Angeles, CA, United States.
2. Man, K., **Chen, Q.**, & Li, J. (2026, April 10). *Assessing pre-knowledge cheating via innovative measures*. NCME 2026 Annual Meeting, Los Angeles, CA, United States.

WORKSHOPS

1. Man, K., Toton, S., Gorney, K., & **Chen, Q.** (2025, April). *Applying Data Mining Methods to Detect Test Fraud* [Workshop]. NCME 2025 Annual Meeting, Denver, CO, United States.
2. Man, K., Toton, S., Gorney, K., Li, J., & **Chen, Q.** (2026, April). *Applying Data Mining in Multi-Agent Systems for Test Fraud Detection* [Workshop]. NCME 2026 Annual Meeting, Los Angeles, CA, United States.

SKILLS

1. Bayesian Inference: Proficient in probabilistic programming and Bayesian modeling using NumPyro, PyMC, Stan, and JAGS; specialized in GPU-accelerated computation for MCMC methods via NumPyro.
2. AI & Machine Learning: Experienced in the development of Multi-Agent Systems (MAS) and reinforcement learning; practical application of supervised/unsupervised learning and neural networks using Python (scikit-learn, PyTorch).

3. Advanced Statistical Modeling: Strong background in Structural Equation Modeling (SEM), Hierarchical Linear Modeling (HLM), and Item Response Theory (IRT) using R (lavaan, mirt, lme4) and Mplus.
4. Data Engineering: Competent in data management and analysis pipelines using SQL, Python (Pandas, NumPy), and R; experience with big data visualization and reproducible research workflows.
5. Collaborative Research: Effective collaborator in multi-disciplinary teams.

WORKING EXPERIENCE

The University of Alabama Aug 2025 - Present
Teaching Assistant Tuscaloosa, AL
Developed tutorial materials and data examples for BER 345: Educational Statistics, supporting undergraduate students in statistical concepts and applied data analysis.

AWARDS AND SCHOLARSHIPS

Graduate Council Fellowship 2024 – 2025
The University of Alabama
A university-level merit fellowship awarded via departmental nomination, providing a full tuition scholarship and a competitive stipend for exceptional research potential.

REVIEWING SERVICE

Frontiers in Psychology (peer-reviewed journal); National Council on Measurement in Education (NCME) Annual Meeting 2025; American Educational Research Association (AERA) Annual Meeting 2026.

PROFESSIONAL MEMBERSHIPS

National Council on Measurement in Education (NCME) 2024 – Present
American Educational Research Association (AERA) 2025 – Present